Buyer Segmentation Based on Machine Learning and Investment Profiling with Intelligence in the Real Estate Market

Project Report | Unified Mentor / Parcl Co. Limited

1. Executive Summary

The study covers buyer segmentation through client demographics, reasons for acquisition, financing metrics, satisfaction rating, and the sale activity of the properties. The datasets include records of 2,000 clients and 10,000 properties. K-Means was run with four clusters alongside sampled comparison of hierarchical clustering. The dashboard shows silhouette score of 0.130 for K-means and 0.116 for hierarchical clustering, these numbers are low, suggesting that the segments overlap with each other, therefore the results should be viewed as exploratory.

2. Problem Statement and Objectives

The project short states that Parcl doesn't have a clear picture of buyer types, motivations behind investment depending on demographics, regional behaviors regarding investment and customer financing. The objectives are to clean and merge the datasets, identify trends, segment the buyers into clusters, compare different methods of clustering, and present the results via interactive dashboard.

3. Data description and preparation

The dataset clients.csv consists of the columns client_id, client_type, first_name, last_name, date_of_birth, gender, country, region, acquisition_purpose, satisfaction_score, loan_applied, and referral_channel. The dataset properties.csv consists of the columns listing_id, tower_number, transaction_date, unit_category, unit_number, floor_area_sqft, sale_price, listing_status, and client_ref. The client_ref field in the properties dataset has 2695 missing values; during the aggregation step no record is associated with any client. The selling prices were converted from currency format text to numeric values. The records of properties that have been sold have been aggregated by client_ref, and then merged with client profiles. The age is calculated as of 12 October 2026.

4. Methodology

Numeric variables, such as age, satisfaction, quantity of sold properties, total purchase amount, average price of purchase, average area, and number of property types were imputed using median and standardized accordingly. On the other hand, categorical variables were filled in using the mode and subjected to one-hot encoding. In K-Means, k was specified as 4, random_state as 42, and n_init as 10. As regards evaluation, the underlying methodology exploits hierarchical clustering with the help of Ward linkage with the replicable sample of 500 records according to runtime requirements.

5. Results

Value provided in the current live dashboard is summarized in the table below. The algorithm generates Cluster IDs and therefore they should not be treated as established buyer categories. Average purchase value is reported without any changes but it is important to check currency beforehand.

Cluster	ID	Average age	Average satisfaction	Property buyers	Loan applicants	Properties sold	Average purchase price
C1	552	22.66	1.45	178	202	1,936	313,841
C2	627	24.91	3.08	181	232	2,295	422,966
C3	49	27.59	3.49	17	16	360	334,389
C4	772	22.85	4.09	239	286	2,714	310,044

Currently available scores: K-Means silhouette 0.130; Hierarchical silhouette 0.116.

Main findings

C3 is the unit with the lowest amount of customers with the size of 49; meanwhile, average C3 satisfaction is recorded as 3.49.

C4 has the highest recorded average of satisfaction of 4.09.

C2 displays the highest average purchase value (422,966), C3 comes second with 334,389, while C1 and C4 are the least with 313,841 and 310,044 respectively.

As the silhouette values obtained are low, these segments cannot be used as clusters.

6. Dashboard Design

The Streamlit dashboard shows the segmentation of buyers, charts for investment behaviour, and other geographic analyses. The sidebar allows the filtering of data according to the country, region, acquisition, and client type. It is important to note that the dashboard is only for exploratory purposes and does not indicate direct investment advice.

7. Recommendations

Validate the four exploratory clusters with business stakeholders

Investigate whether the small C3 segment is an actual segment.

Improve linkage for property records without the “client_ref” value.

Try to derive features that are more effective.

8. Restrictions

Due to the absence of a clear-cut income field in the dataset, the descriptive segments that contain income details or "big money" will not be verifiable. Property connectivity has not been completed for 2,695 rows. Silhouette scores had been derived from sampling and do not serve as formal indicators for model comparison. The report matches the cluster statistics provided in the dashboard image; if the analysis.py file is launched again or the dataset is changed, the report should be generated anew.

9. Summary

The workflow combines client information and sold property activity to demonstrate buyer segmentation, model comparison, and market data analysis. The dashboard shows differences in the age of the buyers, level of satisfaction, and average purchase amount, while low values of silhouette scores mean that there is an issue with overlapping clusters. It is recommended to improve property-client connection, work on features, and conduct business validation before the system can be used.

Reproducibility: Use python analysis.py to generate the clusters again and use streamlit run app.py to run the dashboard.